

Intelligent Material Selection Using Particle Swarm Optimization and Ashby Indices for Aerospace Structures

1 Introduction

Material selection is a cornerstone of advanced structural design, directly governing performance, weight, cost, and reliability. In aerospace applications, this choice is particularly challenging due to competing demands (specific strength, stiffness, thermal resistance, damage tolerance) and the complexity of loading scenarios. While classical methods like Ashby charts and decision matrices are effective, they often require extensive manual iteration. This project develops an automated system that combines a structured material database, mathematical Ashby indices, a constraint-filtering retrieval module, Particle Swarm Optimization (PSO), and a language-model-based coordinator to assist engineers in selecting optimal materials with minimal manual effort.

2 Problem Statement

When designing load-bearing components such as wing spars, ribs, or landing gear, the engineer must select from dozens of candidate alloys (aluminium, titanium, high-strength steels, composites). Two main difficulties arise: (1) hard-constraint filtering – factors like maximum operating temperature or density instantly disqualify many materials, yet manual filtering is time-consuming and error-prone; (2) hidden multi-objective optimisation – although Ashby performance indices guide the choice, identifying the best trade-off under real-world restrictions (corrosion, strain limits) requires intelligent search. Traditional approaches handle these steps separately and rely heavily on human judgement. This project bridges the gap by integrating constraint filtering, index-driven optimisation, and validation into an automated pipeline.

3 Objectives

The project aims to build a software system that:

- Extracts operational requirements from natural-language input (e.g., “working temperature 200 °C, maximum weight saving, damage tolerance”).
- Automatically eliminates unsuitable materials based on hard constraints (maximum service temperature, yield strain, etc.).
- Uses Particle Swarm Optimization to rank the remaining materials according to the most relevant Ashby index.
- Validates the top-three candidates from a practical aerospace perspective (galvanic compatibility, formability).

- Outputs a ranked list of three best materials accompanied by a technical justification report.

4 Methodology

The core of the system rests on two computational pillars: a material database of 30–50 aerospace alloys and composites, each described by properties such as elastic modulus, yield strength, density, maximum working temperature, and fracture toughness, and a set of Ashby performance indices (specific flexural strength, specific stiffness, damage tolerance) implemented as analytical functions. A retrieval-based constraint filter first screens the database against hard limits derived from the design requirements. The surviving materials are then passed to a Particle Swarm Optimizer whose objective function maximises the selected Ashby index, yielding a ranked list of the top candidates.

These two modules—filter and optimiser—are embedded in an automated workflow coordinated by a language model. The coordinator accepts a natural-language description of the design problem, translates it into technical constraints and the appropriate Ashby index, triggers the filter and the PSO, and finally assembles a human-readable engineering report that presents the three best materials along with their performance validation. The entire pipeline operates without manual intervention, ensuring speed and repeatability.

5 Expected Outcomes

- A curated, extendable aerospace material database.
- A retrieval-plus-PSO ranking engine that operates autonomously.
- An automated pipeline that converts user requirements into a technical justification report.
- For each test scenario (wing spar, UAV ribs, landing gear), a report listing the three optimal materials with supporting rationale.
- Well-documented, reusable source code for future research on intelligent material selection.

6 Conclusion

By combining Ashby’s methodology, metaheuristic optimisation, and a language-model-based coordinator, the proposed system automates the entire material selection decision chain. It accelerates the design phase, reduces manual error, and provides engineers with a reliable, repeatable decision-support tool for advanced aerospace structures.